Asian Journal of Philosophy (2025) 4:109
https://doi.org/10.1007/s44204-025-00334-3

ORIGINAL ARTICLE

The dynamics of personal autonomy: a comprehensive framework for evaluating human-carebot interactions

Linus Ta-Lun Huang[1] · **Tzu-Wei Hung**[2] · **Ying-Tung Lin**[3]

Received: 21 September 2024 / Accepted: 15 September 2025
© The Author(s) 2025

Abstract

Carebots are robots developed to assist users within healthcare contexts. The current literature on carebots and their impacts on autonomy has standardly discussed only some components of autonomy rather than offering a comprehensive analysis. In this paper, we propose a more thorough framework. This framework applies Catriona Mackenzie's multidimensional account of autonomy to factors in human–robot interactions that significantly affect personal autonomy. We show, through examining existing carebots with this framework, that their impacts on autonomy are indeed multifaceted. We also argue that their impacts are dynamical in structure, where promoting some conditions of autonomy results in diminishing others or even at the expense of others. One implication of our analysis is that this framework can and should be applied to assess the impacts of carebots upon autonomy at the designing phase, as well as when we consider adopting their use in different contexts.

Keywords Social robotics · Healthcare · Personal autonomy · AI4SG · Bioethics · Human–machine interaction

✉ Ying-Tung Lin
linyingtung@nycu.edu.tw

Linus Ta-Lun Huang
LinusHuang@cuhk.edu.hk

Tzu-Wei Hung
htw@sinica.edu.tw

[1] Public Humanities Program, Department of Cultural and Religious Studies, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong

[2] Institute of European and American Studies, Academia Sinica, No. 128, Sec. 2, Academia Rd., Nankang District, Taipei 115, Taiwan

[3] Institute of Philosophy of Mind and Cognition, National Yang Ming Chiao Tung University, No. 1001, Daxue Rd. East Dist., Hsinchu City 300093, Taiwan

Published online: 17 October 2025

1 Introduction

Carebots are social robots designed to offer cognitive, physical, or social support to care recipients and/or caregivers. These robots play an increasingly important role in healthcare, such as assisting elderly people or people with disabilities (Pu et al., 2019; Sathish et al., 2020), children with autism spectrum disorders (Pennisi et al., 2016), or patients living with dementia (Moyle et al., 2019) and depression (Collins et al., 2018). In the case of in-home senior care, for instance, carebots are utilized to improve elders' activities of daily living (e.g., eating, dressing, toileting, or transportation). Given these uses, carebots are one of the emerging fields in AI that attract large amounts of funding and sustained attention. Notably, although various applications have been proposed recently, most carebots are still prototypes and have not yet led to the commercial launch of a product (Van Aerschot & Parviainen, 2020). However, precisely because the role of carebots is currently marginal, this is the ideal opportunity to consider their ethical issues before this technology is further integrated into society. After all, once the genie is out of the bottle, it is very hard to put it back in (Forge, 2008). In this article, we will focus on an essential topic: the carebots' potential impact on the personal autonomy of care recipients (hereafter, the "user").

Recent studies have attended to the risk that robots pose to personal autonomy, shedding light on how social robots could threaten human autonomy in various conditions. However, many works do not provide an explicit definition or analysis of the concept of autonomy they employ. For example, Sharkey and Sharkey (2012) observe that "loss of freedom is related to the issue of autonomy" (p. 33) and that the autonomy of elderly people increases as their dependence on other people decreases. However, they do not explicitly define what autonomy is nor why a reduction of freedom necessarily undermines autonomy. Similarly, Jenkins and Draper (2015) argue that forceful and intrusive monitoring by carebots may violate elderly users' autonomy. However, they do not seem to explicitly distinguish between different senses of autonomy, mixing the notion of autonomy as competence in one's decision-making capacities with the distinct notion of autonomy as freedom from manipulation.

However, the lack of explicit definition is not the only problem. Even studies that define and discuss different dimensions of autonomy explicitly often oversimplify autonomy and only focus on a subset of the dimensions. For example, Pirhonen et al. (2020) highlight three different senses of autonomy and argue that carebots may strengthen one while simultaneously posing a threat to others. Calvo et al. (2020), by contrast, analyze autonomy into a few key dimensions in light of the concept of the theory of self-determination and present an interesting perspective on how to design AI systems to support personal autonomy. Nevertheless, both works fail to provide a comprehensive analysis of personal autonomy. When researchers or designers omit some aspects of autonomy, the resulting analyses risk overlooking crucial ethical impacts of robots in healthcare settings.

In this article, we will offer a more comprehensive framework ideally suited to examine the impact of carebots on personal autonomy. Section 2 will introduce the first component of this framework: Mackenzie's multidimensional account of personal autonomy, which analyzes autonomy into three dimensions and seven related

conditions. By employing her account, we are not attempting to provide a reductive definition. Rather, we follow her in acknowledging that "autonomy" is complex. Care recipients in healthcare contexts often experience dependence, vulnerability, and social stigma—factors that purely procedural or purely individualistic theories of autonomy can overlook. By employing Mackenzie's approach, we can capture both the *relational* and the *personal* dimensions of users' agency. Section 3 offers the second component of our framework: the key factors in human–robot interactions that are likely to exert significant influence on personal autonomy in the healthcare context. Then, we apply this framework in Sect. 4 to examine how existing carebots, as lab prototypes and commercial products, promote some dimensions of autonomy while simultaneously threatening others. Section 5 illustrates that the important tradeoffs between different conditions of autonomy, as discovered via the application of this framework, can often come about due to the adoption of carebots. As we will show, the promotion of one autonomy condition can lead to the reduction of others in certain cases. Importantly, the analyses in Sects. 4 and 5 indirectly demonstrate the robustness of Mackenzie's framework as we apply it to tangible cases of carebots and uncover complexities that simpler accounts of autonomy might miss. We conclude with some suggestions regarding how this framework can help engineers, policymakers, and healthcare providers make more thorough assessments of the benefits and risks of designing and adopting carebots.

2 Dimensions of personal autonomy

We will begin by presenting Mackenzie's (2014) multidimensional account of personal autonomy, which is particularly suitable for our analysis. Her account identifies a broader set of conditions that undergird both relational and individualistic approaches to autonomy.[1] It also assists our focus on carebots users' vulnerability and dependence on their social environment, which is a particularly salient consideration within the healthcare context. In adopting her account, however, we will remain neutral regarding her commitment to relational autonomy. Instead, we focus on the central features of these dimensions while staying neutral with regard to the subtle but important debates about them.[2] In the following discussion, we introduce

[1] For an influential collection on relational autonomy, see Mackenzie & Stoljar, (2000). The classic individualistic rendering of autonomy can be found in Mill's *On Liberty* (2003).

[2] There are two main debates concerning relational autonomy (Stoljar, 2018). One concerns the difference between causal and constitutive approaches: the causal approach focuses on the effect of external factors, such as one's interpersonal relations, on the relevant dimension involved. The constitutive approach, however, considers that at least some of these external factors serve as the "defining conditions" of the relevant dimension. For example, under the causal approach, certain factors (such as social recognition of an agent's relevant capacities) bring about the capacity for autonomy. Meanwhile, under the constitutive approach, receiving social recognition is *part of being autonomous*. For brevity of exposition, we implicitly adopt the causal approach in this paper for the dimensions of self-governance and self-authorization (see Sects. 2.1 and 2.3 below).

The other key debate concerns the difference between substantive and procedural accounts of autonomy. According to "substantive" theories, autonomy requires agents to possess particular values or beliefs in order to count as autonomous. Meanwhile, "procedural" accounts emphasize the relations between these values and beliefs, where these beliefs or values must be subject to critical reflection without an eye to content. We set aside this consideration in this paper.

three dimensions—self-governance, self-determination, and self-authorization—each of which involves several conditions that can be satisfied to varying degrees.[3] These three dimensions and related conditions should not be read as an exhaustive list; rather, they capture major strands in contemporary analyses of autonomy. Mackenzie's approach consolidates a richly interwoven set of themes into a working model—one we consider invaluable in analyzing robotics in healthcare settings.

2.1 Self-governance: competence and authenticity conditions

Self-governance refers to the skills or capacities necessary to make decisions and produce actions that express one's core beliefs and motives. According to Mackenzie and Stoljar (2000), it involves two broad types of conditions: competence conditions, which focus upon required decision-making capacities, and authenticity conditions, which spell out the conditions under which one's beliefs and motives are truly one's own. We consider each in turn.

Competence conditions refer to the psychological capacities deemed necessary to effectively make and enact decisions given one's goals and contexts. Capacities that are commonly taken as competence conditions include the capacity to understand relevant information; to make reasonable assessments of this information; to reach a decision about what to do in light of one's assessments and one's goals; and the ability to execute decisions. For example, people suffering from serious dementia may have compromised competence conditions for autonomy in certain domains of their lives, such as an inability to consider alternative courses of action.[4] One notable feature of Mackenzie's approach is the acknowledgment that these capacities depend on interpersonal, social, and institutional scaffolding for their development and continuing exercise. As such, different social environments can promote or diminish an agent's competence. For example, according to the "interactionist" theory of reasoning (for an important version, see Mercier & Sperber, 2017), one reasons best within a social context. When we reason as individuals, we tend to be biased and jump to conclusions—we look for evidence in support of our preferred conclusion, including evidence of poor quality. When reasoning is performed in a social context, by contrast, we exchange arguments with our interlocutors and expose the flaws in the arguments with others, with the best ideas winning out. It follows that in a social environment where the exchange of arguments is discouraged or repressed in some

[3] See Mackenzie (2014) for more detail.

[4] Different theories provide characterization deemed as the core features underlying these capacities (e.g., reason-responsiveness) (Fischer & Ravizza, 2000) or demand additional capacities (e.g., critical reflection, imaginative skills, emotional responsiveness, and skills for self-understanding) on top of those commonly enumerated (Mackenzie, 2014).

domains, one's reasoning capacity is likely to be compromised to some degree in those domains.

Authenticity conditions specify the components of one's motivational structure (e.g., desires, values, reasons, intentions) that are genuinely one's own. For instance, if one is hypnotized into forming an intention to perform a chicken dance on stage, this intention is inauthentic as the source of this intention is found in the hypnotist, not the hypnotized. The claim here is that the agent can be alienated from some of their motives due to internal or external factors.[5] Christman (2004), for example, argues that a motive is authentic if, and only if, the agent did not resist, or would not have resisted, its development when attending to the processes of its formation.[6] This again draws attention to how an oppressive social system can affect one's autonomy. For example, sexist societies can socialize people into developing inauthentic motives. In many societies, women are informed that they should avoid fields of study, such as mathematics or physics in favor of fields that more closely align with sexist stereotypes regarding the innate abilities of women, such as the humanities. Were one to reflect upon the process of gender socialization, one would likely find this process of socialization to be oppressive and, as a result, resist the motives developed through this process as unendorsable and inauthentic (Mackenzie & Stoljar, 2000).

2.2 Self-determination: freedom and opportunity conditions

We have discussed the dimension of self-governance. Now, we turn to the dimension of self-determination, which involves having the freedom and opportunity to self-govern in one's social context. The core intuition behind this dimension is that "Agents who stand in relations of subordination, subservience, deference, or economic or psychological dependence, for example, cannot be self-governing because they do not enjoy practical control over their lives" (Mackenzie, 2014, p. 24). Consequently, these two broad types of conditions, *freedom* and *opportunity*, capture the external conditions for autonomy. The distinction between freedom and opportunity conditions, according to Mackenzie, roughly reflects the distinction between freedom from interference versus the substantive freedom to access certain goods.

Freedom conditions specify the social and political constraints or liberties that interfere with or enable self-governance, respectively, in a social context. Some of the commonly recognized political and personal liberties include freedom of thought and expression; freedom of conscience and religion; freedom of movement;

[5] Theories analyze these relations differently. "Structural" accounts analyze them entirely in terms of the relation of a motive with other internal states, such as coherence with the agent's higher-order desires and values, among others (Frankfurt, 1971; Watson, 2004). Here, we follow alternative "historical" approaches that pay attention to internal relations as well as external ones, such as the historical process of practical identity formation within social contexts.

[6] Note, however, that Christman does not consider himself to be adopting a relational autonomy approach, as he seeks to develop a historical account of authenticity.

as well as freedom from coercion, manipulation, exploitation, and violence (Mackenzie, 2014).

Opportunity conditions include those "(personal, social, political) opportunities that need to be available to agents" in order for them to self-govern in social contexts (Mackenzie, 2014, p17). In unpacking this concept of opportunity conditions, we follow Mackenzie in understanding it in terms of capabilities. Capability theories focus on what one can actually choose to do or be in life, such as participating in politics or being a parent, among other options (understood as "functionings"), and one's substantial opportunities ("capabilities") to achieve these functionings (Nussbaum, 2007). Important capabilities include health and nutrition (being able to have good physical/mental health and adequate nutrition), safety (being able to be secure from danger, violent assault, etc.), mobility (being able to move freely), play (being able to laugh and enjoy recreation), education, gender equity, the ability to participate in social cooperation, and the ability to interact with people without stigma, among others (Robeyns & Byskov, 2020). These capabilities, therefore, amount to the material conditions that must be obtained in order for agents to be capable of achieving their goals and a life of value.

2.3 Self-authorization: accountability, self-evaluative attitudes, and social recognition conditions

Even when an agent meets the conditions for self-governance and self-determination, she (or her peers) may still not recognize her autonomy, and this lack of recognition can function to undermine the autonomy of the agent. The self-authorization dimension aims to capture the conditions under which one is regarded by oneself or others as having the "normative authority" to self-govern and self-determine. According to Mackenzie, there are three types of conditions: accountability, self-evaluative attitudes, and social recognition. We explore each in turn.

Accountability conditions concern agents' attitude or disposition to hold themselves responsible, accountable, and answerable to others for their beliefs, values, commitments, actions, and reasons. This dimension reflects the fact that we are under constant demands to explain, justify, and take responsibility for our actions (and the reasons behind them) in social settings. As a result, autonomous action requires that one understands and responds to such demands as well as regards oneself as a valid recipient of these demands.

The condition of self-evaluative attitudes is about an agent's judgments and evaluations of her own thoughts, desires, and behaviors. These judgments and evaluations, dependent upon interpersonal and social relations, are important to maintaining one's own sense of normative authority. This is because reliance on one's self-reflection and judgment is important for autonomy, but such reliance is only possible when one has a sense of self-worth and competence, especially when faced with the kind of social demands discussed above, such as justifying our actions. Some of the important self-evaluative attitudes are self-respect—"to regard oneself as the moral equal of others, as having equal standing to have one's views and claims taken seriously" (Mackenzie, 2014, p. 37); self-trust—the ability to trust one's own

judgment, emotional responses, etc.; and self-esteem—thinking of one's life, goals, and values as meaningful and worthwhile.

Social recognition conditions, finally, reflect the conditions under which other people consider the agent to have normative authority. Normative authority refers to the notion that an individual is understood to have the ultimate say over their lives, such that interventions upon them amount to the imposition of somebody else's will.[7] Social recognition is important for autonomy partly because those self-evaluative attitudes addressed above are dependent on *interpersonal* relations, e.g., whether one's decisions are treated as worthy of respect. A failure of social recognition can thus undermine relevant self-evaluative attitudes.

To illustrate the three conditions above, we turn to the well-known example of the "Deferential Wife" (Hill, 1991). The Deferential Wife thoughtfully and (apparently) willingly prioritizes and defers to her husband's preferences and choices. In fact, she tends not to form her own preferences or choices, and even when she does, she considers them to be less significant than her husband's. It has been argued (Stoljar, 2018) that she is the victim of oppressive circumstances where women are unjustifiably considered occupying a subservient role, regarded as having inferior worth, not capable of rational decision-making, and, ultimately, not a full participant in society (i.e., that the conditions of social recognition are undermined). These social attitudes serve to erode her self-worth, confidence in her own reasoning, and her ability to see herself as an equal to her husband (such that the self-evaluative attitude conditions are threatened). Ultimately, the way she answers for herself in response to demands to justify her choices is governed by the reasons, preferences, and commitments of her husband (i.e., accountability conditions are also diminished). This example demonstrates quite succinctly the benefits of appealing to a multidimensional account, as this allows us to see in some detail exactly how the Deferential Wife's autonomy is actively undermined both by other agents and society, *and* how these factors explain how these features become internalized by the agent herself.

The above amounts to an introduction and demonstration of a multidimensional account of personal autonomy. In what follows, we will analyze the impact of carebots on their users utilizing this multidimensional approach. As we will see through our analysis, these different types of conditions are not only conceptually related to one another, but they are also causally interdependent. By identifying these interrelated conditions, this account will enable a better analysis of the impact of carebots on personal autonomy.

3 Key factors in human–robot interaction

The previous section served to analyze personal autonomy's multidimensionality. In this section, we provide an analysis of the other primary aspect of our framework: the key factors of human-carebot interactions that are likely to significantly impact

[7] See Mackenzie (2008) for a related discussion.

personal autonomy. This will allow us in the next section to analyze the impacts upon autonomy that can come about via the use of carebots.

We first distinguish three types of carebots based on their function. We break down these functions into the following broad types:

> *Cognitive assistance robots* (hereafter: CARs) are designed to promote the user's cognitive and emotional capacities, such as memory, decision-making, and the capacity to understand the emotional states of oneself or others.[8]
> *Physical assistance robots* (PARs) are designed to promote motor functions, such as standing up, moving, and grabbing objects.
> *Social assistance robots* (SARs) are designed to promote the user's social capacities, such as communication, social interactions, and cooperation.

[9]

For more detail regarding our taxonomy of carebot functions, see Table 1.

We now turn to the key factors in human–robot interaction (HRI), which describe features of carebot usage and design that are useful for analyzing the effects of carebots upon different conditions of autonomy. HRI analysis also aids in clarifying the way tradeoffs between these conditions come about (as we will demonstrate in Sects. 4 and 5). Note that these factors are not to be understood as independent, as they often influence each other in various ways.

The *relational role* refers to the role carebots play in their specific interactions with humans, including assisting, training, or automating various functions; monitoring activities; and providing companionship. As robotic service cannot be completed without participation by the user, the role of robots is necessarily relational. For example, to support independent living at home, a carebot can *assist* the user in feeding themselves; *automate* a motor task, such as bringing objects to the user; *train* the user's memory through cognitive training programs; *monitor* their movement for fall prevention; or provide companionship by engaging with the user.

Technology refers to the different technical processes, methods, or artifacts used to implement the functions of carebots, such as via Machine Learning (ML), Big Data, the Internet of Things (IoTs), smart sensors, analytics, and so on.[10] These

[8] For instance, AMIGO has tablet-based interactive apps integrated into its chest to mitigate the effects of dementia. The apps are in the style of serious games that playfully support multimodal training (e.g., improving the user's working memory). The user's data are collected via the tablet and analyzed with a recommender engine to facilitate more accurate and customized training (Paletta et al., 2019).

[9] For example, Scassellati et al.'s (2018) SAR promotes the communication skills of children with ASD. They integrate a robot, a touchscreen monitor, and two cameras to support triadic interactions between the robot, the child, and the caregiver. Their robot can make eye contact and share attention with children. It also offers feedback to and guides the children in interactive games. Caregivers' reports and their robotic data both indicate less prompting and increased communication in these children.

[10] Machine Learning (ML) depends on the development of mathematical theories and algorithms that allow computers to recognize complex patterns in examples or data; Big Data refers to the mass collection and generation of information that can be analyzed to reveal patterns, associations, or other useful information; the Internet of Things (IoTs) refers to the relatively recent phenomena of applying web connectivity to artifacts, to improve the functionality or usability of those artifacts; analytics is the systematic computational analysis of data and is used to discover, interpret, and communicate significant patterns in data (Sharda et al., 2020).

Table 1 This table shows the main functions and examples of current carebots in each type. Laboratory prototypes are indicated with an asterisk (*) to distinguish them from commercially available products

	Cognitive assistance robot (CAR)		Physical assistance robot (PAR)		Social assistance robot (SAR)	
Main functions	· Emotion monitoring · Emotional support (e.g., pet-like robots for animal therapy) · Entertainment (e.g., games, acting as a music player) · Cognitive coaching and stimulation (e.g., multimodal cognitive enhancement therapy) · Reminders (e.g., scheduling, medicine reminders) · Notification (e.g., news and weather) · Cognitive behavior therapy · Telepresence and telemedicine		· Action, body, and environment monitoring · Physical rehabilitation and stimulation · Fall alert and fall prevention · Feeding and object manipulation · Motion assistance (e.g., wearable robots) and acting as a walking helper (for elders and those living with disabilities) · Telepresence and emergency call · Lifting/transferring between beds and wheelchairs · At-home self-care (often integrated with smart home devices or IoT etc.) · Body surrogacy		· Increase motivation for social interaction (robot-based therapy for ASD) · Help ASD children learn distrust and deception detection	
Instances	Mobiserv Matilda* NAO Pepper PARO Care-O-bot 4 Hobbit*	Woebot +me* AMIGO*	ROBEAR* JUVA* Hobbit* Mobiserv Obi NAO Pepper	Care-O-bot 4 (when equipped with arms) +me* iGo HIMICO* PR2	NAO Pepper AMIGO* Matilda* Care-O-bot 4 +me* Hobbit*	Woebot

different technologies facilitate various kinds of interactions between humans and carebots. For instance, ML can enable carebots to determine the preferences of a particular user (e.g., that they are vegetarian) and predict the user's behavior to provide relevant information (suggesting dietary options).[11] These technologies thereby serve to personalize carebots to meet a specific user's needs. While these technologies are not yet fully utilized in currently available products, future carebots will increasingly incorporate them.

Individualization refers to the degree to which features of carebots (such as their user interface, the algorithms at work, or the information provided to users) are fine-tuned for a particular user. Individualization, implemented through technologies such as ML and Big Data, could even allow for persistent or real-time fine-tuning. This exemplifies how one HRI factor can influence another.

Behavior shaping refers to the different types of strategies that are intentionally designed to influence a user's behavior. Some key strategies (Schmidt & Engelen, 2020) include the following:

(1) *Coercion* involves the removal of options from the user. When a website forces you to watch an advertisement before directing you to the content you wish to consume (thereby removing the option for directly consuming the content), it has adopted a subtle coercion strategy.
(2) *Incentivizing* changes the reward structures of the different options, such as when a health initiative pays people to quit smoking.
(3) *Nudging* predictably alters the user's behaviors by changing the ways options are presented to them without changing the options themselves. For example, by adopting an opt-out system for organ donation, we can expect an increase in the number of donors, as we can depend upon a general tendency to simply accept the default option.
(4) *Explicit reasoning strategies* aim at people's behavior by primarily engaging in explicit reasoning, such as via rational persuasion, highlighting relevant information, suggesting available options, and so on.

The strategies and the extent to which carebots restrict and shape the agent's behavior can impact different dimensions of autonomy and lead to a tradeoff between them (as we will see in Sect. 5).

Dependence refers to the degree to which a user relies on carebots to perform a given task. Dependence can result from external conditions (such as a lack of alternative available assistance from others) or due to the condition of the user (as in the case of limited cognitive or physical capacities and coping flexibility, or due to relying upon a carebot for an extensive period of time). For instance, a user with compromised cognitive or physical functionality is more likely to depend on carebots

[11] ML predictions rely on previous data, so issues in the input or training data can lead to problems such as discrimination, bias, and privacy violations (Huang et al., 2022; Hung and Yen 2023; Lin et al., 2020; Yen and Hung 2021).

than healthy individuals as they (unlike those who are entirely self-reliant) require assistance to perform certain tasks (such as taking medicine on time).

Balancing interests refers to the varying interests of stakeholders which are embodied in the design of carebots. Carebots are designed, for the most part, to meet the user's interests; however, other interests, such as those of the manufacturers and those of broader society, can also be embodied in their design. For instance, carebots could be designed to promote sponsored healthy products by presenting targeted advertising to the user.

Transparency refers to the degree to which the goals and means of carebots functionality are accessible and understandable by their user. For example, when a carebot suggests to a user different options of music to play it may not be transparent to the user how these songs are chosen out of a larger set of her favorites (demonstrating "means non-transparency"). Moreover, the user may also be unclear as to why carebots suggest listening to music at a particular moment ("goal non-transparency").

As we will see in the next two sections, the multidimensional account of personal autonomy, together with the key factors in HRI, will provide a useful framework for analyzing carebots' impact on personal autonomy, as well as the way tradeoffs occur between components of autonomy.

4 Evaluating the current impact of carebots

In this section, we apply our framework to analyze the impact of *currently available* carebots because they offer concrete examples of how machines can interact with humans who need care. This analysis is not intended to be an exhaustive account and will instead focus on only those impacts we consider to be most significant. We examine, through the key factors of HRI (as discussed above), how different conditions for autonomy are promoted or diminished. We will demonstrate that, while current carebots do positively affect *some* conditions of autonomy, they can also threaten others.[12]

Our goal is not to provide an argument in favor of carebots or against them. Even though our treatment will consider a greater number of examples where carebots promote the conditions of autonomy than threaten it, this is partly explained due to the promotion of autonomy being one of the explicit goals of carebot designers. However, as we will show, carebots are entirely capable of undermining the autonomy of their users, given the possibility of overlooking unintended consequences or oversights in design. In fact, we should note that for every case of promotion of conditions of autonomy discussed here, we could potentially also find a corresponding case that undermines the same condition. Furthermore, the promotion of one condition can often result in compromising another (In Sect. 5, we will highlight

[12] We will put aside the issue of informed consent in this paper, as it is a complex and controversial issue but not particularly germane to our discussion of the application of carebots. See Veatch (1995) and Beauchamp and Childress (2001) for two central texts on informed consent.

some important tradeoffs between autonomy conditions.) Finally, due to the scope of this paper, we do not consider the probabilities of these positive and negative impacts. Therefore, our analysis serves to examine the overall impact of carebots on autonomy.

4.1 Impacts of current cognitive assistance robots

Current CARs directly affect the conditions of competence, opportunity and self-evaluative attitudes positively, and freedom conditions negatively. We elaborate on each of these in turn.

First, competence conditions could be improved by current CARs by augmenting human decision-making capacities through the application of different *relational roles* and *behavioral shaping* strategies. First of all, CARs can assist a user's decision-making by providing relevant, structured information, such as a reminder to take medicine (Fischinger et al., 2016; Görer et al., 2017). Such reminders improve the user's ability to consider relevant information and, as a result, marginally improve their competence conditions. In addition, CARs, through cognitive training, can also help the user restore some of their cognitive capacities. AMIGO, for instance, is a carebot that provides multimodal training for dementia patients (Paletta et al., 2019). It utilizes tablet-based interactive games to restore a user's cognitive capacities, such as working memory—a crucial capacity underlying complex decision-making.

Second, several opportunity conditions are promoted to some extent by current CARs, such as those of health and nutrition, safety, play, and so on. For example, through different applications of *relational* roles and facilitated by *technology* such as smart sensors for tactile, vision, posture, and audition, the pet-like robot PARO can engage in some social interaction with its user to provide companionship (such as entertainment and emotional support) and cognitive-behavioral training. It benefits mental health (e.g., reducing stress, Aminuddin et al., 2016) as well as aiding the cognitive capacities of dementia patients (e.g., visual and verbal interaction; Šabanović et al., 2013).[13] As such, it promotes the opportunity conditions of health (in particular, mental health) and play (being able to enjoy recreation).

Third, as for the *self-evaluative attitudes* conditions, current CARs demonstrably promote these conditions by preventing depressive states, which often negatively impact one's self-esteem and self-respect. For example, Woebot is a self-care chatbot that can be installed on smart mobile devices. Studies suggest that this conversational chatbot can assist and train its user in self-monitoring their negative thought patterns (e.g., perfectionism, self-defeating thinking). It can then employ different *behavioral shaping* strategies similar to cognitive-behavioral therapy to suggest or nudge the user to adopt alternative interpretations and framings of the user's

[13] Despite evidence suggesting that animal therapy benefits the mental health of those affected by PTSD (Dell et al., 2018; Scotland-Coogan et al., 2020), further research is needed to establish the therapeutic effect of a pet-like robot in various contexts.

experiences and, as a result, prevent such thought patterns from worsening their emotional state (Monnier, 2020).

Despite these apparent benefits to personal autonomy, it is not all good news. Current CARs (as well as PARs and SARs) have demonstrated negative impacts on freedom conditions. Some carebots need to collect the user's personal data in order to perform their function (Demirci, 2018; Fitzpatrick et al., 2017; Paletta et al., 2019).[14] Even when such data are collected legally and with the user's consent, it can nevertheless lead to a variety of issues related to privacy: a breach of which threatens to undermine freedom conditions. For example, Woebot has to collect users' responses for analysis, including sensitive and personal data (e.g., sexual orientation or political views). This may threaten privacy and cause psychological pressures, leading to self-censorship, thereby risking undermining freedom of expression.[15] When PARO collects biometric data in order to provide entertainment and social support to its user, the user may, again reasonably, worry that this information is now available to be used to influence their behavior in ways that infringe on their rights to freedom from manipulation and exploitation. When PARs monitor their users' movements for concern as to their safety (see Sect. 4.2), the user could also believe that their freedom of movement is compromised. Note that we are not saying that these data *are*, in fact, being used to undermine the user's freedom in these ways, but the *mere fear* that they *could* be is enough to undermine these relevant freedom conditions to some degree. By analogy, modern CCTV surveillance polices human behavior not because the individual *is* monitored, but because they *could be* monitored, identified, and punished. Moreover, the lack of *transparency* (at least from the user's perspective) as to why carebots collect and utilize certain personal data can further drive the conflict between these privacy-related concerns and freedom. Of course, the user also cannot rule out the possibility that these data are, in fact, being used to undermine various freedom conditions, such as freedom from manipulation (consider the Facebook–Cambridge Analytica data scandal), due to the commercial interests of the carebot company.

In conclusion, while current CARs can directly affect the competence, opportunity, and self-evaluative attitudes conditions positively and freedom conditions negatively, they have no significant direct impacts on authenticity conditions, as well as the accountability and social recognition conditions, as far as we are able to ascertain. Nevertheless, we want to note that conditions in the dimension of

[14] There may not be clear differences between collecting, storing, and processing data. For example, real-time monitoring carebots for fall prevention may simultaneously receive and process data from environmental changes and the user's bodily movement; where the information may be temporarily stored in a cache instead of in a hard disk, which could make these differences blur. Given that privacy breaches can already occur at the stage of data *collection*, we will focus on this sense of privacy here.

[15] Fear or causing fear is a threat to freedom of expression. For example, Tibetan activist Chemi Lhamo and Taiwanese transgender Leon Tsai were attacked by Chinese nationalists when they ran for student president and vice-president at the University of Toronto in 2019 (Laurent, 2020). Also, digital platforms and content providers are not always trustable. Facebook engaged in country-wide censorship of content in Vietnam (Amnesty International, 2020), and YouTube deleted comments with keywords insulting the Chinese Communist Party (Vincent, 2020). In 2017, Cambridge University Press (CUP) also removed over three hundred articles that may anger China from the publisher's website (Carrico, 2018).

self-authorization (i.e., accountability, self-evaluative attitudes, and social recognition) can often be affected *indirectly* (i.e., as mediated through other autonomy conditions) by all three types of carebots we have identified. For instance, as one's competence and opportunity increase, these can promote one's sense of accountability and self-evaluative attitudes (self-respect, self-esteem, self-trust, and so on) as well as one's social recognition (i.e., other people come to see the user as more autonomous). On the other hand, it is also possible that a more comprehensive adoption of carebots in the future can negatively impact self-authorization. We come back to this issue in Sect. 5.5.

4.2 Impacts of current physical assistance robots

Currently, PARs can directly promote opportunity conditions, but they also can undermine freedom conditions. For example, PARs need to monitor and analyze personal sensitive data (e.g., biometrics and activities of daily living) of the user to offer customized service. The more data it can access, the greater the risk of breaching the users' privacy. Just as in the case of CARs, this data gathering could threaten privacy and freedom of political expression. Furthermore, PARs can indirectly (through improving opportunity conditions) promote conditions in the dimension of self-authorization. While PARs' influences are similar to CARs in impacting upon the freedom conditions and those conditions contained within the self-authorization dimension (see discussions in Sect. 4.1), we will focus on how PARs can promote opportunity conditions.

Current PARs have positive impacts on several *opportunity conditions*, such as mobility, health and nutrition, safety, and the ability to participate in social cooperation through various *relational roles* and *technology*. For example, ROBEAR, an experimental carebot which has a bear-like appearance, promotes mobility: it can lift a user and transfer them between beds and wheelchairs (Szondy, 2018). Obi promotes health and nutrition when it assists a physically disabled user in feeding themselves via the use of its robotic arms (Grace, 2016). Similarly, iGo is a walking assistant robot integrated with the user's body and utilizes motion detection technology to maintain the user's balance. It helps its users navigate the environment and enables them to participate better in social activities (Abou Allaban et al., 2020). Finally, PR2 employs the technology of augmented reality (AR) to enable body surrogates. It is designed to help users with physical impairments complete daily tasks independently, such as teeth brushing and shaving. Studies suggest that PR2 improves the quality of life of patients with profound motor deficits (Grice & Kemp, 2019; Kapusta & Kemp, 2019). In addition, PR2 has often been adapted by its users for novel functions. A user, for example, controlled PR2 to hold a hairbrush to scratch his head and a towel to wipe his mouth—functions not foreseen by its designers. As put by the user himself: "it completely obviated the need for a human caregiver once the robot was turned on" (Grice & Kemp, 2019, p. 21).

To summarize, while current PARs directly affect opportunity conditions positively and freedom conditions negatively, they can also impact conditions in the dimensions of self-authorization indirectly. With various opportunity conditions

improved, the user can live a more self-reliant and independent life and is likely to experience some improvement in self-evaluative attitudes (self-esteem and self-respect in particular), social recognition, and accountability conditions. However, while PAR can indirectly promote self-evaluative attitude conditions through improved opportunity conditions, with increasing *dependence* on PARs instead of human care, in the long run, there could be a tradeoff between the opportunity and self-authorization dimensions. We will return to this point in Sect. 5.5.

4.3 Impacts of current social assistance robots

Current SARs can directly promote opportunity conditions and diminish freedom conditions (as we have seen in Sect. 4.1). They can also indirectly (through improving opportunity conditions) promote conditions in the dimension of self-authorization.

Current SARs, in utilizing technology to promote a user's *relational roles*, can promote the opportunity conditions of health (in particular, mental health) and the ability to participate in society. For instance, a wearable carebot called me+ has biometric sensors that gather physiological data to provide information about the mental states of children with ASD. This information is used by a therapist to better engage the child in training to improve therapeutic outcomes, such as better social skills (Ozcan et al., 2016). With improved social function as a result, the user's relevant opportunity conditions are promoted.

Similarly, with improved opportunity conditions related to one's social capabilities, the user is likely to have improved self-esteem and self-respect (self-evaluative attitude conditions). These benefits are likely to result in others being more likely to recognize them as autonomous agents (i.e., through promoting the social recognition condition). However, there is still the worry that human interactions or human care will be increasingly replaced by SARs due to a tenuous assumption that SARs can provide the kinds of social engagement humans need in healthcare. In Sect. 5.5, we will discuss how *overdependence* on carebots can undermine the dimension of self-authorization.

In conclusion, we have here evaluated the current carebots' impacts on different conditions of autonomy (see Table 2 for a summary), demonstrating that the impacts of carebots on autonomy are multifaceted and that different carebots promote and undermine different conditions of autonomy in different contexts and to differing degrees.

5 The dynamics of personal autonomy

We have demonstrated in the previous section how MacKenzie's analysis of autonomy can be applied to analyze the impacts of current carebots upon different conditions of personal autonomy. In this section, we focus on the dynamics *between* conditions in human-carebot interaction. We will show, through selective illustrations, that in the promotion of one condition of autonomy, carebots may inadvertently

Table 2 This table summarizes the impacts of current carebots upon the components of autonomy

Condition\type		Cognitive assistance robot (CAR)	Physical assistance robot (PAR)	Social assistance robot (SAR)
Self-governance	Competence	Promoted	No significant direct impacts	No significant direct impacts
	Authenticity	No significant direct impacts	No significant direct impacts	No significant direct impacts
Self-determination	Freedom	Diminished	Diminished	Diminished
	Opportunity	Promoted	Promoted	Promoted
Self-authorization	Accountability	Promoted indirectly (via increased competence and opportunity conditions)	Promoted indirectly (via increased opportunity conditions)	Promoted indirectly (via increased opportunity conditions)
	Self-evaluative attitudes	Promoted		
	Social recognition	Promoted indirectly (via increased competence and opportunity conditions)		

reduce others; in addition, such tradeoffs between promotion and reduction of autonomy conditions can be hard, though not impossible, to avoid.

5.1 Authenticity vs. freedom tradeoff

We start with a plausible case of how future carebots can promote authenticity conditions while reducing freedom conditions. We have discussed how a current CAR, Woebot, through engaging its user in simple cognitive-behavioral therapy, can promote self-evaluative attitude conditions (Sect. 4.1). A possible future development on such CARs—let us call it Woebot 2.0—could potentially promote the user's authenticity conditions through more sophisticated psychological counseling. For example, Woebot 2.0 might engage the user to reflect on their beliefs, values, and intentions and thereby help clarify what motives are most authentic to them and constitute their core self.

However, for Woebot 2.0 to achieve such autonomy promotion, it may employ various new *technologies* to individualize and fine-tune psychological counseling sessions through continuously collecting personal information and analyzing it in ways that are quite distinct from the method employed by human counselors. For example, natural language processing and computer vision (including facial recognition) can be applied to capture subtle changes in the user's tone and facial expression and infer useful information for the purpose of therapy. Big data could also uncover the user's digital footprint across different sources, such as social media and private communications, for the purposes of further fine-tuning. These data could be utilized for the purpose of personal informatics for users to gain a deeper understanding of their own cognitive processes and behaviors; for instance, it can keep the user aware of when there may be the influence of cognitive biases or coercion (Li et al., 2010).

So in order to better serve the *interests* of the user, Woebot 2.0 could potentially gain access to a large amount of intimate information about the user previously only accessible in circumscribed and independent contexts (e.g., information usually covered in a counselor's report, multiple disparate digital devices, social media platforms, and email correspondence). There is not much doubt that this brings a new potential for the promotion of authenticity conditions. However, given that the commercial *interests* of companies can often play a problematic role in how these data are utilized beyond the original, consented to, service, it can seriously undermine freedom conditions through an increased threat to the user's privacy, inevitably raising serious concerns about freedom of expression, freedom from manipulation and exploitation, freedom of movement, and beyond (as discussed in Sect. 4.1).[16]

We should stress that this tradeoff can be ameliorated to various degrees. The collection, storage, and processing of personal data can be protected through both better design and the creation and application of legal frameworks/institutions (High-Level Expert Group on Artificial Intelligence, 2019). However, given how poorly

[16] We can also foresee similar tradeoffs occurring during the promotion of competence and opportunity conditions via the collecting of personal data for individuating carebot services.

privacy issues are often handled currently, we should be extremely cautious when considering these challenges. Accordingly, whether it is desirable to allow carebots to have access to their user's personal data in such an extensive way raises ethical considerations that require further investigation.

5.2 Competence vs. freedom tradeoff

Through different strategies of *behavioral shaping*, carebots can promote the user's competence conditions, as we have seen in Sect. 4. In this and the next section, we focus on likely tradeoffs with freedom and authenticity conditions. We will first look specifically at how carebots may utilize nudging to promote competence while also undermining freedom.[17] CARs can promote the user's competence conditions, for example, when they are used to nudge a user, who needs and wants to improve their diet, to make healthier decisions more often by presenting the healthiest option as a more salient option or by visualizing nutrition data in relation to the context of the user's diet. This kind of nudging, which helps structure the user's decision environment, amounts to a straightforward action in line with the agent's values and goals, and thereby a *justified* action as it improves their capacity to act according to these values and goals.[18] Despite this apparent benefit, however, the user's freedom conditions may also be undermined, as we will now explore.

The first issue is with freedom from manipulation. Due to the lack of *transparency* (as the user is often unaware of how nudges influence their behavior), there is a sense that the user is subject to outside control. So nudges can be seen as an intentional manipulation of a user, and this can hold even in the case where the goals of the manufacturers or designers of carebots are aligned with those of the user. To make things worse, a user suffering from cognitive deficits—exactly the kinds of users carebots are intended to help—may be even less likely to recognize nudges and their influences, compounding their susceptibility to these manipulative effects. Indeed, with emerging *technological* innovations and the resulting *individualization* of carebots to particular users, freedom from manipulation may be undermined to an even greater degree. For example, when a company collects user data and applies machine learning to individualize the nudges applied to a specific user, it can make the carebot function more effectively in shaping the behavior of users. Finally, the

[17] Other strategies, such as incentivizing, can also have similar tradeoffs, but we refrain from discussing them here.

[18] Some may argue that autonomy involves rationality, but nudging makes use of the automatic process (System 1) and bypasses rational capacities (System 2); thus, nudging cannot undermine autonomy. However, first, human rationality is not confined to System 2 processing, and System 2 can affect System 1 as well. Both systems make use of cognitive resources, but System 2 requires working memory, and System 1 does not (Van Gestel et al., 2020). Studies show that some nudging relies on cognitive processes; for example, alarm sounds for seatbelts and traffic light labels for nutrients on packaging help scaffold the slower decision-making processes (Hansen & Jespersen, 2013; Schmidt & Engelen, 2020). Second, deliberative processes can inhibit the effectiveness of nudging (Van Gestel et al., 2020), but if a care receiver's working memory is impaired, nudging may conflict with the care receiver's freedom from being manipulated. Thus, these carebot developments make it less likely that it is the actual user's values and goals (as opposed to some third parties) that will be promoted.

dependence of users upon carebots can result in a larger impact of robot nudges due to the amount of interactions the user has with carebots, over lengthy timeframes. The more embedded a carebot is in a user's life, the greater the danger that their freedom will be imperiled by their influences.

There is an additional issue when we consider freedom from exploitation. Different *interests* involved in designing nudges mean that, for example, they can serve the *corporate interests* of those who create these carebots in addition to or, crucially, at *the expense of* user interests. The user may be nudged to eat a heartier diet by being presented with a salient option of a particular brand—a brand that pays fees to the carebot company for the privilege. In such cases, the agent is potentially being exploited for the financial gains of a private corporation.

In short, a tradeoff between competence and freedom conditions exists when we consider the impacts of carebots. This tradeoff is, however, not inevitable. We may be able to avoid manipulation by making nudges transparent to and controllable by the user. We may also be able to avoid exploitation by ensuring that nudge design does not benefit any third party in an unethical manner. However, these are certainly challenging topics both in theory (for example, figuring out what types of transparency and control are needed to prevent manipulation and what counts as benefits in an appropriate way) and in practice (establishing regulatory frameworks and institutions to ensure compliance).

5.3 Competence vs. authenticity tradeoff

We now turn to how carebots can, in the process of promoting competence, undermine authenticity. We focus on the least controversial strategy of behavioral shaping: the targeting of explicit reasoning, such as via notification, suggestion, and rational persuasion of users. For example, CARs can promote a senior user's competence conditions by reminding them to take a rest from physically strenuous activities, suggesting recreational activities that are suitable for their level of cognitive and physical capacities, or rationally persuading users to avoid engaging in unsafe behaviors.

Nevertheless, carebot algorithms, developed through the *technology* of ML from Big Data (among other methods), may mirror societal biases and form part of an oppressive social system (Barocas et al., 2017; Huang et al., 2024; Prince & Schwarcz, 2019; Williams et al., 2018, Sechman & Huang, 2025). This may result in the user making more competent decisions while, at the same time, rendering users less capable of making authentic decisions, or making users more likely to develop inauthentic motives. For example, CARs may embody ageism, which refers to discriminatory prejudices, stereotypes, and practices against older people. Ageist attitudes include beliefs that older people are less capable cognitively (assuming the elderly necessarily have worse memories, increased inflexibility of thought, and posing a danger in cognitively demanding situations, such as driving in traffic) and less capable physically (needing constant rest, poor recovery after injury, vulnerability in physically demanding environments). Such CARs may direct elderly users to information and options that they would not otherwise consider (due to having authentic

values and desires, say, to "have a life" and ranking stimulating and adventurous experiences over safety and longevity). In practice, a CAR might frequently draw the user's attention to safety considerations, even in a relatively safe environment. They may also only suggest options for recreational activities that are low risk and require less cognitive and physical ability. Thus, such CARs—the use of which will likely carry authority in the healthcare context—could prevent the user from forming intentions and goals to engage in activities that could be otherwise rewarding and in line with their authentic values; moreover, they could serve to "discipline" the user via rational persuasion to form habits and values that conform to ageism in the long term (Morley & Floridi, 2019). Were the user to reflect on these problematic processes, they may not find these motives or decisions to be endorsable and authentic.

Moreover, the lack of *transparency* of how the algorithms work (as it is unlikely that the user will understand how and why the carebot offers the information and options it does) can make it harder for them to become aware of their oppressive power. This problem is made worse by machine learning, which can produce "black boxes" such that even the designers cannot explain (or predict) why an algorithm behaves as it does (Castelvecchi, 2016; Papernot et al., 2017; Samek et al., 2017).

Again, such tradeoffs between competence and authenticity are not an empirical necessity. It seems possible to eliminate some problematic biases from the carebot algorithms and even deliberately design "good biases" in the coding of carebots that can help actively fight against, say ageism. Yet it will require both significant empirical and institutional innovations in order to prevent such tradeoffs from continuing to have a likely negative effect on user autonomy.

5.4 Opportunity vs. competence tradeoff

This section and the next focus on the tradeoffs resulting from the HRI factor *dependence*. Here, we show that while CARs can promote opportunity conditions, users' long-term dependence on CARs may threaten competence conditions in certain situations.

As discussed in Sect. 4.1, CARs can promote various opportunity conditions. For instance, a common problem faced by healthcare clinicians is non-adherence with treatment regimes, usually typified by failing to take required medicines. This problem is explicable by reference to varying cognitive factors (for example, patients living with cognitive or mental disabilities or other cognitive deficits such as memory loss). These cognitive deficits lead to impaired ability to plan, organize, and execute tasks required to adhere to formulated treatment plans. To improve compliance with treatment plans, CARs, through the *relational role of* assisting, can help manage treatment compliance by, say, audibly reminding users both when to take medicines and what medicines to take. As such, this kind of CAR helps improve the opportunity condition of health.

However, a possible outcome is a decrease in competence conditions as a result of cognitive *dependence* on this type of assistance. For example, a user may persistently rely on the assistance of CARs across different domains and tasks that would

otherwise require exercising their cognitive capacities, including those involved in competence conditions. Over a long period of time, this can result in diminished competence conditions. Whether this concern is warranted will depend on the availability of more direct empirical evidence; nevertheless, some indirect evidence exists. For instance, in aging research, it has been demonstrated that cognitive inactivity is associated with reduced cognitive performance, whereas persons with mentally demanding lifestyles show higher performance in intellectual flexibility, memory, verbal ability, and so on (Kramer et al., 2004; Salthouse, 2006; Williams & Kemper, 2010).

Again, this tradeoff between opportunity and competence conditions is not an empirical necessity. For example, CARs can support medication adherence through *training* of cognitive functions, which allows CARs to promote both opportunity and competence conditions at the same time (Williams & Kemper, 2010). As a result, when we consider the adoption and design of carebots, we should be especially attentive to the unintended consequences of different methods of promoting a particular condition of autonomy to improve the user's overall autonomy, as it could come at the expense of others.

5.5 Opportunity vs. self-authorization (accountability, self-evaluative attitudes, and social recognition) tradeoff

In this final section, we will discuss the second tradeoff associated with the HRI factor of *dependence*. As aging populations, as well as populations of those living with disabilities, put increasing pressure on scarce human resources in healthcare, we can predict a trend to utilize carebots in place of human health professionals whenever relevant technology makes it possible. Note, however, that increasing dependence on carebots to deliver care in various domains, despite its benefits in promoting various opportunity conditions, poses threats to the self-authorization dimension of autonomy.[19]

At the center of this tradeoff is a concern regarding so-called *deep care*. This can be defined as "care as feeling and as reciprocity of feeling" (Coeckelbergh, 2010, p. 183). Despite the fact that carebots can be designed to offer cognitive, physical, and social assistance to users by responding to their behaviors and demands in ways similar to human care, it is often argued that the care they provide is comparatively shallow (Coeckelbergh, 2010). This is because carebots lack the real emotional engagement and meaningful communication that, as Sparrow and Sparrow (2006) argue, are critical for genuine care.

A significant lack of deep care could negatively impact the dimension of self-authorization. To maintain the conditions of self-evaluative attitudes, the agent needs to live in a healthy social environment where regular and genuine social interactions nurture their sense of self-worth, self-respect, and self-trust. Genuine social interactions are also crucial for maintaining one's social recognition conditions,

[19] Some of these robots have been trialed already. For example, Robear, the robot that looks like a bear. It is used for carrying patients and providing a sympathetic face.

which requires the recognition of one's normative authority for self-governing and self-determination by other *agents*. Finally, the maintenance of accountability conditions requires the continuing exercise of one's disposition to hold oneself accountable to other agents. The worry, thus, is that carebots may replace the deep care provided by either healthcare professionals or the user's family and friends with a pale imitation—imitation care. This will mean the user could live in an increasingly impoverished social environment which will diminish the conditions related to self-authorization over time.

The remedy for such a tradeoff is, of course, to supplement, not substitute, deep care afforded by humans with the imitation care provided by carebots, thus ensuring users continue to be socially active and have their emotional and social needs fulfilled. However, it is worth noting that this may be an overly optimistic view, given current trends in automation in general. Artifacts—especially artifacts that aim to be integrated into everyday life—are hard to control or predict, and *successful* artifacts can have destabilizing effects within capitalistic employment markets. Automation, as a general phenomenon, has already resulted in large-scale unemployment within many fields (sometimes dubbed "technological unemployment"), and without top-down efforts t*o prevent* the replacement of human agents capable of providing deep care with carebots, there seems to be a real danger that this eventuality will come to pass (Luke, 2019). This is especially salient when we consider that care work is often understood as "menial" and often undertaken by those with fewer qualifications or insecure standing in communities, such as immigrants. As a result, this kind of work is more susceptible to disruption via technologies, such as carebots.

In this section, our goal was reasonably modest: to show that promoting one condition of autonomy can inadvertently come at the expense of others. For our purposes, this brings into relief two key findings. First, it is unhelpful to discuss enhancing or promoting autonomy without spending the time to enumerate *what aspect* of autonomy is being targeted for promotion. Without unpacking the effects of carebots and the *way* they bring about these effects, one can easily fall into the trap of thinking that the effects are an unalloyed good (or an unalloyed bad). Instead, the truth is that carebots (like many artifacts) represent a mixed blessing, requiring careful consideration by designers and regulators. Second, this analysis reveals that designers or engineers really are obligated to think through the *complete* effects of their designs, and not end their considerations once they have found that a particular carebot will have a positive impact on some component of autonomy. Instead, it seems as if there is a *prima facie* case here that consideration should be paid to the unintended effects that will come about through the use and widespread adoption of these artifacts (Forge, 2008).

6 Conclusion

We have provided an in-depth framework for examining the impact of carebots on personal autonomy. This framework analyzes the concept of autonomy into three distinct dimensions: self-governance (competence and authenticity conditions), self-determination (freedom and opportunity conditions), and self-authorization

(accountability, self-evaluative attitudes, and social recognition conditions). It also identifies seven key HRI factors that significantly impact personal autonomy in human–robot interactions: relational roles, technology, individualization, behavioral shaping, dependence, balancing interests, and transparency.

Through applying this framework to examine current and future carebots, we have shown that the impacts of carebots upon personal autonomy are multifaceted: different carebots can promote and diminish different conditions of autonomy in different contexts and to differing extents. Moreover, we have also illustrated how tradeoffs between conditions of autonomy can often be brought about via the use of carebots, as they often promote one condition of autonomy at the expense of others. Even though such tradeoffs are not an empirical necessity, they present considerable challenges when we aim to promote the overall autonomy of the user. In fact, we have demonstrated that in analyzing the impact of carebots upon personal autonomy, it is wise to avoid making overgeneralizing claims, such as the claim that carebots straightforwardly violate user autonomy or that carebots straightforwardly promote autonomy. Instead, it is necessary to apply a comprehensive framework to specific cases because they will inevitably involve different human–robot interaction factors and dynamic tradeoffs among different conditions. This analysis represents a framework that should be of use to robot engineers and designers, policymakers, and healthcare providers in determining whether the design or introduction of carebots in a given context is worth these tradeoffs and provides guidance as to the ethical and prudential costs and benefits of such a move. This will allow decision-makers to isolate key considerations that should factor in any thorough analysis of the impacts of these new artifacts. Finally, our analysis illustrates how the multidimensional framework reveals overlooked ethical tradeoffs and helps guide designers and policymakers. In this way, our discussion serves, in part, as a vindication of the nuanced distinctions Mackenzie draws.

Acknowledgements We thank Pei-hua Huang and Anson Fehross for their help in improving the earlier versions of the manuscript.

Author contribution LH, TH, and YL contributed equally to this work.

Funding Open Access funding enabled and organized by National Yang Ming Chiao Tung University. Dr. Linus Ta-Lun Huang is supported in part by grant GRF-16600823 from Hong Kong SAR Research Grants Council. Dr. Tzu-Wei Hung is supported by National Science and Technology Council, Taiwan (112–2410-H-001–031-MY3). Dr. Ying-Tung Lin is supported by a Taiwan National Science and Technology Council (NSTC) project (113–2410-H-A49 -042 -MY3) and by an NSTC grant under the ERA-NET NEURON (Network of European Funding for Neuroscience Research) scheme (113–2923-H-A49-002-MY3).

Data availability We do not analyze or generate any datasets because our work proceeds within a theoretical approach.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Abou Allaban, A., Wang, M., & Padır, T. (2020). A systematic review of robotics research in support of in-home care for older adults. *Information (Basel), 11*(2), Article 75.

Aminuddin, R., Sharkey, A. J., & Levita, L. (2016). Interaction with the Paro robot may reduce psychophysiological stress responses. In *2016 11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (pp. 593–594). https://doi.org/10.1109/HRI.2016.7451872

Barocas, S., Bradley, E., Honavar, V. G., & Provost, F. J. (2017). Big data, data science, and civil rights. *ArXiv, abs/1706.03102*. https://doi.org/10.48550/arXiv.1706.03102

Beauchamp, T., & Childress, A. E. (2001). *Principles of biomedical ethics* (5th ed.). Oxford University Press.

Calvo, R. A., Peters, D., Vold, K., Ryan, R. M., Burr, C., & Floridi, L. (2020). Supporting human autonomy in AI systems: A framework for ethical enquiry. In C. Burr & L. Floridi (Eds.), *Ethics of digital well-being. philosophical studies series* (Vol. 140). Springer. https://doi.org/10.1007/978-3-030-50585-1_2

Carrico, K. (2018). China studies between censorship and self-censorship. *Made in China Journal, 3*(2), 12–15. https://www.chinoiresie.info/china-studies-between-censorship-and-self-censorship/

Castelvecchi, D. (2016). Can we open the black box of AI? *Nature News, 538*(7623), 20–23.

Christman, J. (2004). Relational autonomy, liberal individualism, and the social constitution of selves. *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition, 117*(1/2), 143–164.

Coeckelbergh, M. (2010). Health care, capabilities, and AI assistive technologies. *Ethical Theory and Moral Practice, 13*(2), 181–190. https://doi.org/10.1007/s10677-009-9186-2

Collins, S., Ŝabanović, S., Fraune, M., Randall, N., Eldridge, L., Piatt, J. A., et al. (2018, March). Sensing companions: Potential clinical uses of robot sensor data for home care of older adults with depression. In *HRI '18: Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 89–90). https://doi.org/10.1145/3173386.3177047

Demirci, H. M. (2018). *User experience over time with conversational agents: Case study of Woebot on supporting subjective well-being*. Master's thesis, Middle East Technical University (Turkey).

Fischer, J. M., & Ravizza, M. (2000). *Responsibility and control: A theory of moral responsibility*. Cambridge University Press.

Fischinger, D., Einramhof, P., Papoutsakis, K., Wohlkinger, W., Mayer, P., Panek, P., Hofmann, S., Koertner, T., Weiss, A., Argyros, A., & Vincze, M. (2016). Hobbit, a care robot supporting independent living at home: First prototype and lessons learned. *Robotics And Autonomous Systems, 75*, 60–78.

Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): A randomized controlled trial. *JMIR Mental Health, 4*(2), Article e19.

Forge, J. (2008). *The responsible scientist*. Pittsburgh University Press.

Frankfurt, H. G. (1971). Freedom of the will and the concept of a person. *The Journal of Philosophy, 68*(1), 5.

Görer, B., Salah, A. A., & Akın, H. L. (2017). An autonomous robotic exercise tutor for elderly people. *Autonomous Robots, 41*(3), 657–678.

Grace, R. (2016). Obi feeds more than the imagination: A decade in the making, this self-dining robot offers a taste of freedom & independence to millions of disabled. *Plastics Engineering, 72*(10), 18–22.

Grice, P. M., & Kemp, C. C. (2019). In-home and remote use of robotic body surrogates by people with profound motor deficits. *PLoS One, 14*(3), Article e0212904.

High-Level Expert Group on Artificial Intelligence. (2019). *Ethics guidelines for trustworthy AI*. European Commission. https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

Hill, T. (1991). *Autonomy and self-respect*. Cambridge University Press.

Huang, L. T.-L., Chen, H.-Y., Lin, Y.-T., Huang, T.-R., & Hung, T.-W. (2022). Ameliorating algorithmic bias, or why explainable AI needs feminist philosophy. *Feminist Philosophy Quarterly, 8*(3/4). https://doi.org/10.5206/fpq/2022.3/4.14347

Huang, L.T.-L., Papyshev, G., & Wong, J. K. (2024). Democratizing value alignment: From authoritarian to democratic AI ethics. *AI and Ethics*. https://doi.org/10.1007/s43681-024-00624-1

Hung, T.-W., & Yen, C.-P. (2023). Predictive policing and algorithmic fairness. *Synthese*. https://doi.org/10.1007/s11229-023-04189-0

Jenkins, S., & Draper, H. (2015). Care, monitoring, and companionship: Views on care robots from older people and their carers. *International Journal of Social Robotics, 7*(5), 673–683.

Kapusta, A., & Kemp, C. C. (2019). Task-centric optimization of configurations for assistive robots. *Autonomous Robots, 43*(8), 2033–2054.

Kramer, A. F., Bherer, L., Colcombe, S. J., Dong, W., & Greenough, W. T. (2004). Environmental influences on cognitive and brain plasticity during aging. *The Journals of Gerontology. Series A, Biological Sciences and Medical Sciences, 59*(9), M940–M957.

Laurent, M. (2020). Filial nationalism in global competition: The 2001 reform of Mandarin textbooks. *Pacific Affairs, 93*(3), 543–566.

Li, I., Dey, A., & Forlizzi, J. (2010). A stage-based model of personal informatics systems. In *Proceedings of the SIGCHI conference on human factors in computing systems (CHI '10)* (pp. 557–566). Association for Computing Machinery. https://doi.org/10.1145/1753326.1753409

Lin, Y.-T., Hung, T.-W., & Huang, T. L. (2020). Engineering equity: How AI can help reduce the harm of implicit bias. *Philosophy & Technology*. https://doi.org/10.1007/s13347-020-00406-7

Luke, T. W. (2019). Afterword: On education and technological unemployment. In M. Peters, P. Jandrić, & A. Means (Eds.), *Education and technological unemployment* (pp. 343–349). Springer. https://doi.org/10.1007/978-981-13-6225-5_22

Mackenzie, C. (2008). "Relational autonomy, normative authority and perfectionism." *Journal of Social Philosophy, 39*(4), 512–533.

Mackenzie, C. (2014). Three dimensions of autonomy: A relational analysis. In A. Veltman & M. Piper (Eds.), *Autonomy, oppression and gender* (pp. 15–41). (Studies in feminist philosophy). Oxford University Press.

Mackenzie, C., & Stoljar, N. (Eds.). (2000). *Relational autonomy: Feminist perspectives on autonomy, agency, and the social self*. Oxford University Press.

Mercier, H., & Sperber, D. (2017). *The enigma of reason*. Harvard University Press.

Mill, J. S. (2003). In D. Bromwich & G. Kateb (Eds.), *On liberty*. Yale University Press.

Monnier, D. (2020). Woebot: A continuation of and an end to psychotherapy? *Psychotherapies, 40*(2), 71–78.

Morley, J., & Floridi, L. (2019). Enabling digital health companionship is better than empowerment. *The Lancet Digital Health, 1*(4), e155–e156. https://doi.org/10.1016/S2589-7500(19)30079-2

Moyle, W., Jones, C., Murfield, J., Thalib, L., Beattie, E., Shum, D., & Draper, B. (2019). Using a therapeutic companion robot for dementia symptoms in long-term care: Reflections from a cluster-RCT. *Aging & Mental Health, 23*(3), 329–336.

Nussbaum, M. (2007). Human rights and human capabilities. *Harvard Human Rights Journal, 20*, 21.

Paletta, L., Dini, A., Murko, C., Yahyanejad, S., & Augsdörfer, U. (2019, June). Estimation of situation awareness score and performance using eye and head gaze for human-robot collaboration. In *Proceedings of the 11th ACM symposium on eye tracking research & applications* (pp. 1–3). https://doi.org/10.1145/3314111.3322504

Papernot, N., McDaniel, P., Goodfellow, I., Jha, S., Celikand, Z. B., & Swami, A. (2017). Practical black-box attacks against machine learning. In *Proceedings of the 2017 ACM on Asia conference on computer and communications security* (pp. 506–519). ACM. https://doi.org/10.1145/3052973.3053009

Pennisi, P., Tonacci, A., Tartarisco, G., Billeci, L., Ruta, L., Gangemi, S., & Pioggia, G. (2016). Autism and social robotics: A systematic review. *Autism Research, 9*(2), 165–183.

Pirhonen, J., Melkas, H., Laitinen, A., & Pekkarinen, S. (2020). Could robots strengthen the sense of autonomy of older people residing in assisted living facilities?—A future-oriented study. *Ethics and Information Technology, 22*(2), 151–162.

Prince, A., & Schwarcz, D. (2019). Proxy discrimination in the age of artificial intelligence and big data. *Iowa Law Review, 105*, 1257–1318.

Pu, L., Moyle, W., Jones, C., & Todorovic, M. (2019). The effectiveness of social robots for older adults: A systematic review and meta-analysis of randomized controlled studies. *The Gerontologist, 59*(1), e37–e51.

Robeyns, Ingrid; Byskov, Morten Fibieger, "The Capability Approach", In Edward N. Zalta (ed.) *The Stanford Encyclopedia of Philosophy* (Winter 2020 Edition), from https://plato.stanford.edu/archives/win2020/entries/capability-approach/

Šabanović, S., Bennett, C. C., Chang, W. L., & Huber, L. (2013). PARO robot affects diverse interaction modalities in group sensory therapy for older adults with dementia. In 2013 IEEE 13th international conference on rehabilitation robotics (ICORR) (pp. 1–6). IEEE.

Salthouse, T. A. (2006). Mental exercise and mental aging: Evaluating the validity of the "use it or lose it" hypothesis. *Perspectives on Psychological Science, 1*(1), 68–87.

Samek, W., Wiegand, T., & Müller, K.-R. (2017). Explainable artificial intelligence: Understanding, visualizing and interpreting deep learning models. arXiv preprint http://arxiv.org/abs/1708.08296

Sathish, B. S., Ranganayakulu, A., Revanth, M., & Rao, M. N. (2020). A Novel Design of Service Robot for Aged and Handicapped Using Raspberry Pi. In *5th International Conference on Next Generation Computing Technologies (NGCT-2019)*, Available at SSRN: https://ssrn.com/abstract=3545094 or http://dx.doi.org/10.2139/ssrn.3545094

Schmidt, A. T., & Engelen, B. (2020). The ethics of nudging: An overview. *Philosophy Compass*. https://doi.org/10.1111/phc3.12658

Sechman, M. S., & Huang, L. T. L. (2025). Towards a Mechanistic Account of Embodied Implicit Bias: Heterarchical Control Network and Its Implications for Intervention. *Topoi,* 1-16. https://doi.org/10.1007/s11245-025-10176-6

Sharda, R., Delen, D., & Turban, E. (2020). *Analytics, data science, & artificial intelligence: Systems for decision support*. Pearson Education Limited.

Sharkey, A., & Sharkey, N. (2012). Granny and the robots: Ethical issues in robot care for the elderly. *Ethics and Information Technology, 14*(1), 27–40.

Sparrow, R., & Sparrow, L. (2006). In the hands of machines? The future of aged care. *Minds and Machines, 16*(2), 141–161.

Stoljar, N. (2018). Feminist perspectives on autonomy. In N. Z. Edward (Ed.)*The Stanford Encyclopedia of Philosophy* (Winter 2018 ed.), retreived 31 Jan 2012 from https://plato.stanford.edu/archives/win2018/entries/feminism-autonomy/

Szondy, D. (2015, February 26). *Robear robot care bear designed to serve japan's aging population*. Retrieved July 19, 2018, from https://newatlas.com/robear-riken/36219/

Van Aerschot, L., & Parviainen, J. (2020). Robots responding to care needs? A multitasking care robot pursued for 25 years, available products offer simple entertainment and instrumental assistance. *Ethics and Information Technology, 22*, 247–256.

Van Gestel, L. C., Adriaanse, M. A., & De Ridder, D. T. D. (2020). Do nudges make use of automatic processing? Unraveling the effects of a default nudge under type 1 and type 2 processing. *Comprehensive Results in Social Psychology*. https://doi.org/10.1080/23743603.2020.1808456

Veatch, R. M. (1995). "Abandoning informed consent." *Hastings Center Report, 25*(2), 5–12.

Watson, G. (2004). *Agency and answerability*. Oxford University Press.

Williams, B. A., Brooks, C. F., & Shmargad, Y. (2018). How algorithms discriminate based on data they lack: Challenges, solutions, and policy implications. *Journal Of Information Policy, 8*, 78–115.

Williams, K. N., & Kemper, S. (2010). Interventions to reduce cognitive decline in aging. *Journal of Psychosocial Nursing and Mental Health Services, 48*(5), 42–51.

Yen, C.-P., & Hung, T.-W. (2021). Achieving equity with predictive policing algorithms: A social safety net perspective. *Science and Engineering Ethics*. https://doi.org/10.1007/s11948-021-00312-x

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.